

AuthiChain Vercel Environment Variables Setup Guide

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Purpose: Complete guide for configuring all environment variables in Vercel for production deployment

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Pre-Deployment Checklist

Before setting up Vercel environment variables:

- [] **Stripe Account:** Switch from TEST mode to LIVE mode in Stripe Dashboard
 - [] **Database:** Ensure production PostgreSQL database is provisioned and accessible
 - [] **Domain:** Have your production domain ready (e.g., authichain.com)
 - [] **NFT.Storage:** Verify API key is active and has sufficient storage
 - [] **Cloudflare:** Confirm R2 storage bucket is created and accessible
 - [] **Blockchain:** Ensure you have a secure production wallet private key
 - [] **Security Keys:** Generate strong, unique secrets for production
-

Security Best Practices

Critical Security Guidelines

1. **NEVER commit `.env` files to Git**
 - Add `.env*` to `.gitignore` (already configured)
 - Never share environment variables in chat, email, or documentation
2. **Use Strong Secrets**
 - Generate cryptographically secure random strings for secrets
 - Minimum 32 characters for `NEXTAUTH_SECRET`
 - Use different secrets for each environment (dev, staging, production)

3. Rotate Keys Regularly

- Stripe API keys: Review quarterly
- JWT secrets: Rotate annually
- Database credentials: Update as needed

4. Limit Access

- Only grant Vercel project access to necessary team members
- Use Vercel's role-based access control
- Enable 2FA on all service accounts (Stripe, Vercel, etc.)

5. Environment-Specific Values

- Use different Stripe keys for test vs. production
- Separate databases for development and production
- Different webhook secrets for each environment



Vercel Setup Instructions

Step 1: Access Vercel Project Settings

1. Log into [Vercel Dashboard](https://vercel.com) (<https://vercel.com>)
2. Navigate to your AuthiChain project
3. Click **Settings** → **Environment Variables**

Step 2: Environment Selection

For each variable, select the appropriate environments:

- **Production:** Live environment (required for all variables)
- **Preview:** Staging/preview deployments (optional, can use test keys)
- **Development:** Local development (optional, usually uses `.env.local`)

Recommendation: Set all variables for Production. Use TEST Stripe keys for Preview.

Step 3: Adding Variables

Two methods to add variables:

Method A: Individual Entry (Recommended for sensitive keys)

1. Click "Add New"
2. Enter variable name (e.g., `DATABASE_URL`)
3. Enter value
4. Select environments
5. Click "Save"

Method B: Bulk Import (Faster, but less secure)

1. Click "Add New" → "Paste .env"
2. Copy-paste your environment variables
3. Review each variable carefully
4. Select environments
5. Click "Save"



Warning: Method B is faster but review each variable before saving!

Environment Variables by Category

1 Database Configuration (REQUIRED)

Variable	Description	Security Level	Example Format
<code>DATABASE_URL</code>	PostgreSQL connection string	 CRITICAL	<code>postgresql://user:password@host:5432/dbname</code>

Production Setup:

```
DATABASE_URL="postgresql://prod_user:SECURE_PASSWORD@prod-db-host.com:5432/authichain_prod?connect_timeout=15"
```

Notes:

-  Use SSL connections: Add `?sslmode=require` for production
 -  Enable connection pooling for better performance
 -  Never use the same database for dev and production
-

2 NextAuth Configuration (REQUIRED)

Variable	Description	Security Level	Value
<code>NEXTAUTH_SECRET</code>	JWT signing secret	 CRITICAL	Generate new for production
<code>NEXTAUTH_URL</code>	Your production URL	 IMPORTANT	<code>https://authichain.com</code>

Production Setup:

```
# Generate a new secret (run in terminal):
openssl rand -base64 32

# Then set:
NEXTAUTH_SECRET="YOUR_GENERATED_SECRET_HERE"
NEXTAUTH_URL="https://authichain.com"
```

Notes:

-  Must be HTTPS in production (no http://)
 -  Include subdomain if using one (e.g., `app.authichain.com`)
 -  Never reuse the development secret
-

3 Stripe Configuration (REQUIRED)

⚠ CRITICAL: Switch to LIVE mode in Stripe before production!

A. API Keys

Variable	Description	Security Level	Where to Find
<code>STRIPE_SECRET_KEY</code>	Stripe API secret key	● CRITICAL	Stripe Dashboard → Developers → API Keys
<code>NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY</code>	Stripe public key	● PUBLIC	Stripe Dashboard → Developers → API Keys
<code>STRIPE_WEBHOOK_SECRET</code>	Webhook signing secret	● CRITICAL	Stripe Dashboard → Developers → Webhooks

Getting LIVE Keys from Stripe:

1. Log into [Stripe Dashboard](https://dashboard.stripe.com) (<https://dashboard.stripe.com>)
2. Toggle **View test data** to OFF (enable LIVE mode)
3. Go to **Developers → API Keys**
4. Copy **Secret key** (starts with `sk_live_`)
5. Copy **Publishable key** (starts with `pk_live_`)

Production Setup:

```
STRIPE_SECRET_KEY="sk_live_YOUR_LIVE_SECRET_KEY"
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY="pk_live_YOUR_LIVE_PUBLISHABLE_KEY"
STRIPE_WEBHOOK_SECRET="whsec_YOUR_WEBHOOK_SECRET"
```

B. Product & Price IDs

You'll need to create products in LIVE mode and get new price IDs.

Variable	Description	Tier	Monthly/Annual
STRIPE_CREATOR_PRICE_ID	Creator tier	\$19/mo	Default
STRIPE_CREATOR_MONTHLY_PRICE_ID	Creator monthly	\$19/mo	Monthly
STRIPE_CREATOR_ANNUAL_PRICE_ID	Creator annual	\$190/yr	Annual
STRIPE_PRO_PRICE_ID	Pro tier	\$49/mo	Default
STRIPE_PRO_MONTHLY_PRICE_ID	Pro monthly	\$49/mo	Monthly
STRIPE_PRO_ANNUAL_PRICE_ID	Pro annual	\$490/yr	Annual
STRIPE_ENTERPRISE_PRICE_ID	Enterprise tier	\$199/mo	Default
STRIPE_ENTERPRISE_MONTHLY_PRICE_ID	Enterprise monthly	\$199/mo	Monthly
STRIPE_ENTERPRISE_ANNUAL_PRICE_ID	Enterprise annual	\$1990/yr	Annual
STRIPE_AGENCY_PRICE_ID	Agency tier	\$499/mo	Default
STRIPE_AGENCY_MONTHLY_PRICE_ID	Agency monthly	\$499/mo	Monthly
STRIPE_AGENCY_ANNUAL_PRICE_ID	Agency annual	\$4990/yr	Annual
STRIPE_BRAND_PRICE_ID	Brand tier	Custom	Custom

Creating Products in LIVE Mode:

1. Switch to LIVE mode in Stripe Dashboard
2. Go to **Products** → **Add Product**
3. For each tier (Creator, Pro, Enterprise, Agency):
 - Add product name, description
 - Create two prices: Monthly and Annual
 - Copy each `price_id` (starts with `price_`)
4. Set in Vercel

Production Example:

```

STRIPE_CREATOR_MONTHLY_PRICE_ID="price_1XXXXXXXXXXXXXXXXXXXXX"
STRIPE_CREATOR_ANNUAL_PRICE_ID="price_1YYYYYYYYYYYYYYYYYYYY"
# ... repeat for all tiers

```

C. Product IDs

Variable	Description
STRIPE_EXPLORER_PRODUCT_ID	Explorer tier product ID
STRIPE_CREATOR_PRODUCT_ID	Creator tier product ID
STRIPE_PRO_PRODUCT_ID	Pro tier product ID
STRIPE_ENTERPRISE_PRODUCT_ID	Enterprise tier product ID
STRIPE_AGENCY_PRODUCT_ID	Agency tier product ID

Note: Product IDs start with `prod_`

4 Blockchain & Smart Contract (REQUIRED)

Variable	Description	Security Level
PRIVATE_KEY	Wallet private key	 CRITICAL
POLYGON_AMOY_RPC_URL	Polygon testnet RPC	 PUBLIC
CONTRACT_ADDRESS	Deployed contract address	 PUBLIC
POLYGONSCAN_API_KEY	Polygonscan API key	 IMPORTANT

Production Setup:**⚠️ CRITICAL SECURITY WARNING:**

- Create a NEW wallet for production
- Never use a wallet that holds significant personal funds
- Use a hardware wallet or secure key management system
- Fund the wallet with only what's needed for operations

```

PRIVATE_KEY="0xYOUR_PRODUCTION_WALLET_PRIVATE_KEY"
POLYGON_AMOY_RPC_URL="https://rpc-amoy.polygon.technology"
CONTRACT_ADDRESS="0xYOUR_DEPLOYED_CONTRACT_ADDRESS"
POLYGONSCAN_API_KEY="YOUR_POLYGONSCAN_API_KEY"

```

For Mainnet (Polygon):

```
POLYGON_MAINNET_RPC_URL="https://polygon-rpc.com"
# Or use Alchemy/Infura for better reliability:
# POLYGON_MAINNET_RPC_URL="https://polygon-mainnet.g.alchemy.com/v2/YOUR_KEY"
```

5 NFT.Storage Configuration (REQUIRED)

Variable	Description	Security Level
NFT_STORAGE_API_KEY	IPFS storage API key	● CRITICAL

Production Setup:

1. Log into [NFT.Storage](https://nft.storage) (<https://nft.storage>)
2. Go to **API Keys**
3. Create a new key for production
4. Copy the key

```
NFT_STORAGE_API_KEY="YOUR_PRODUCTION_API_KEY"
```

Notes:

- ☒ Create separate keys for dev, staging, and production
- ☒ Monitor storage usage and limits
- ☒ Enable notifications for storage limits

6 Cloudflare R2 Storage (REQUIRED)

Variable	Description	Security Level
CLOUDFLARE_API_TOKEN	Cloudflare API token	● CRITICAL
CLOUDFLARE_ACCOUNT_ID	Cloudflare account ID	● IMPORTANT
R2_API_TOKEN	R2 storage token	● CRITICAL
R2_ACCESS_KEY_ID	R2 access key	● CRITICAL
R2_SECRET_ACCESS_KEY	R2 secret key	● CRITICAL
R2_ENDPOINT	R2 storage endpoint	● IMPORTANT

Production Setup:

1. Log into [Cloudflare Dashboard](https://dash.cloudflare.com) (<https://dash.cloudflare.com>)
2. Go to **R2** → Create bucket for production
3. Go to **Manage R2 API Tokens**

4. Create a new token with appropriate permissions
5. Copy credentials

```

CLOUDFLARE_API_TOKEN="YOUR_CLOUDFLARE_TOKEN"
CLOUDFLARE_ACCOUNT_ID="YOUR_ACCOUNT_ID"
R2_API_TOKEN="YOUR_R2_TOKEN"
R2_ACCESS_KEY_ID="YOUR_ACCESS_KEY"
R2_SECRET_ACCESS_KEY="YOUR_SECRET_KEY"
R2_ENDPOINT="https://YOUR_ACCOUNT_ID.r2.cloudflarestorage.com"

```

7 Application Configuration (REQUIRED)

Variable	Description	Value
NEXT_PUBLIC_SITE_URL	Public site URL	https://authichain.com
NODE_ENV	Environment mode	production

Production Setup:

```

NEXT_PUBLIC_SITE_URL="https://authichain.com"
NODE_ENV="production"

```

Notes:

- ☒ Must match your actual production domain
- ☒ Use HTTPS in production
- ☒ Don't include trailing slash

8 Feature Flags (OPTIONAL)

Variable	Description	Default
FEATURE_WALLET_AUTH	Enable wallet authentication	true
FEATURE_CRYPTOPAYMENTS	Enable crypto payments	true
FEATURE_AUCTIONS	Enable auction features	true
FEATURE_COLLECTIONS	Enable NFT collections	true

Production Setup:

```

FEATURE_WALLET_AUTH="true"
FEATURE_CRYPTOPAYMENTS="true"
FEATURE_AUCTIONS="true"
FEATURE_COLLECTIONS="true"

```


Notes:

- Set to "false" to disable features
- Useful for gradual rollout or maintenance

Testing & Validation

After Setting All Variables

1. Redeploy Application

```
bash
# Trigger a new deployment in Vercel
git push origin main
```

2. Check Build Logs

- Go to Vercel Dashboard → Deployments
- Check latest deployment logs
- Look for environment variable errors

3. Test Critical Features

- [] Database connection
- [] User authentication
- [] Stripe payment flows
- [] NFT minting
- [] File uploads
- [] Blockchain transactions

4. Monitor Application

- Check Vercel Analytics
- Monitor error logs
- Test all subscription tiers
- Verify webhook deliveries in Stripe

Troubleshooting

Common Issues

Issue 1: Database Connection Failed

Symptoms: App can't connect to database

Solutions:

-  Verify `DATABASE_URL` is correct
-  Check if production database allows connections from Vercel IPs
-  Add `?sslmode=require` to connection string if needed
-  Test connection from Vercel Functions

Issue 2: Stripe Webhook Failures

Symptoms: Webhooks not being received or verified

Solutions:

-  Verify `STRIPE_WEBHOOK_SECRET` matches Stripe Dashboard
-  Check webhook URL is `https://authichain.com/api/webhooks/stripe`

- ☒ Ensure webhook is created in LIVE mode, not TEST mode
- ☒ Test webhook delivery in Stripe Dashboard

Issue 3: NFT Upload Failures

Symptoms: Can't upload NFT images or metadata

Solutions:

- ☒ Verify `NFT_STORAGE_API_KEY` is valid and not expired
- ☒ Check storage quota in NFT.Storage dashboard
- ☒ Verify `R2_*` credentials are correct
- ☒ Test upload endpoint independently

Issue 4: Authentication Not Working

Symptoms: Users can't log in

Solutions:

- ☒ Verify `NEXTAUTH_SECRET` is set and correct
- ☒ Check `NEXTAUTH_URL` matches your production domain (HTTPS)
- ☒ Clear browser cookies and try again
- ☒ Check Vercel logs for NextAuth errors

Issue 5: Blockchain Transaction Failures

Symptoms: Can't mint NFTs or interact with blockchain

Solutions:

- ☒ Verify `PRIVATE_KEY` is correct and wallet has funds
- ☒ Check `CONTRACT_ADDRESS` is the correct deployed contract
- ☒ Verify RPC URL is responsive
- ☒ Check Polygonscan for contract verification

Support & Resources

Stripe Resources

- [Stripe Dashboard](https://dashboard.stripe.com) (https://dashboard.stripe.com)
- [Stripe API Docs](https://stripe.com/docs/api) (https://stripe.com/docs/api)
- [Test vs Live Mode](https://stripe.com/docs/keys#test-live-modes) (https://stripe.com/docs/keys#test-live-modes)

Vercel Resources

- [Vercel Dashboard](https://vercel.com/dashboard) (https://vercel.com/dashboard)
- [Environment Variables Docs](https://vercel.com/docs/environment-variables) (https://vercel.com/docs/environment-variables)
- [Deployment Docs](https://vercel.com/docs/deployments/overview) (https://vercel.com/docs/deployments/overview)

AuthiChain Resources

- Main Documentation: `/authichain_premium/README.md`
- Deployment Guide: `/authichain_premium/DEPLOYMENT_VERCEL.md`
- Security Hardening: `/authichain_premium/SECURITY_HARDENING_REPORT.md` (coming soon)

Security Reminder

Before going live:

- [] All secrets are unique and strong
 - [] All Stripe keys are in LIVE mode
 - [] Production database is separate from dev/test
 - [] Webhook secrets are configured correctly
 - [] Private keys are securely stored
 - [] 2FA is enabled on all service accounts
 - [] Team members have appropriate access levels
 - [] `.env` file is in `.gitignore`
 - [] No secrets are committed to Git
-

Quick Copy Template

Use this template to fill in your values, then paste into Vercel:

```

# Database
DATABASE_URL="postgresql://..."

# NextAuth
NEXTAUTH_SECRET="..."
NEXTAUTH_URL="https://authichain.com"

# Stripe API Keys
STRIPE_SECRET_KEY="sk_live_..."
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY="pk_live_..."
STRIPE_WEBHOOK_SECRET="whsec_..."

# Stripe Price IDs
STRIPE_CREATOR_MONTHLY_PRICE_ID="price_..."
STRIPE_CREATOR_ANNUAL_PRICE_ID="price_..."
STRIPE_PRO_MONTHLY_PRICE_ID="price_..."
STRIPE_PRO_ANNUAL_PRICE_ID="price_..."
STRIPE_ENTERPRISE_MONTHLY_PRICE_ID="price_..."
STRIPE_ENTERPRISE_ANNUAL_PRICE_ID="price_..."
STRIPE_AGENCY_MONTHLY_PRICE_ID="price_..."
STRIPE_AGENCY_ANNUAL_PRICE_ID="price_..."

# Stripe Product IDs
STRIPE_EXPLORER_PRODUCT_ID="prod_..."
STRIPE_CREATOR_PRODUCT_ID="prod_..."
STRIPE_PRO_PRODUCT_ID="prod_..."
STRIPE_ENTERPRISE_PRODUCT_ID="prod_..."
STRIPE_AGENCY_PRODUCT_ID="prod_..."

# Blockchain
PRIVATE_KEY="0x..."
POLYGON_AMOY_RPC_URL="https://rpc-amoy.polygon.technology"
CONTRACT_ADDRESS="0x..."
POLYGONSCAN_API_KEY="..."

# NFT.Storage
NFT_STORAGE_API_KEY="..."

# Cloudflare
CLOUDFLARE_API_TOKEN="..."
CLOUDFLARE_ACCOUNT_ID="..."
R2_API_TOKEN="..."
R2_ACCESS_KEY_ID="..."
R2_SECRET_ACCESS_KEY="..."
R2_ENDPOINT="https://..."

# Application
NEXT_PUBLIC_SITE_URL="https://authichain.com"
NODE_ENV="production"

# Feature Flags
FEATURE_WALLET_AUTH="true"
FEATURE_CRYPTOPAYMENTS="true"
FEATURE_AUCTIONS="true"
FEATURE_COLLECTIONS="true"

```

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